



COMPUTER
ENGINEERING

QUEZON CITY

COURSE CODE/TITLE: CPE 212 Computer Networks
2

SECTION: CPE31S2

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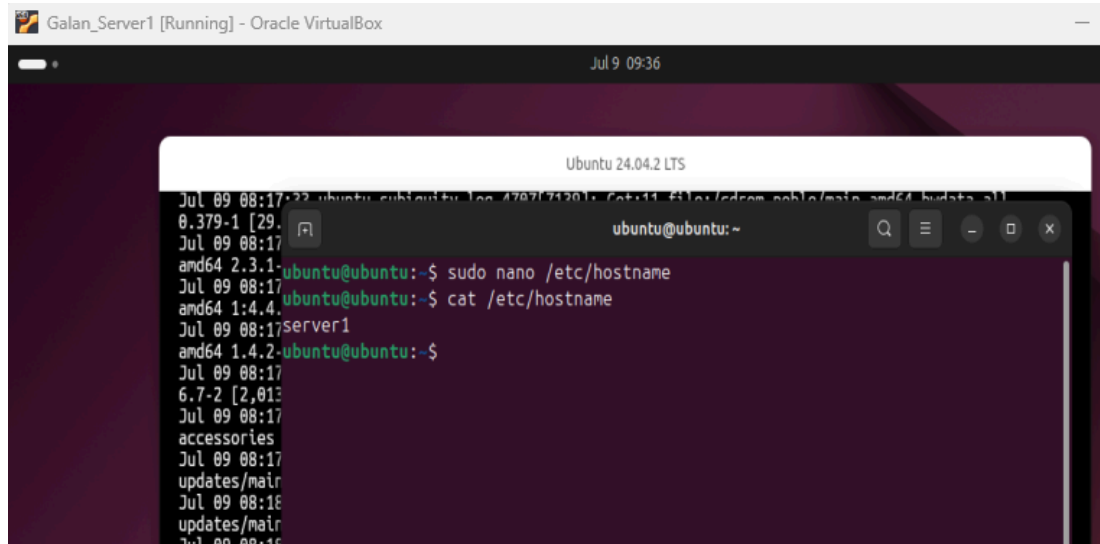
ACTIVITY NO. 1 **Configure Network using Virtual Machines**

1. PROCEDURE OUTPUT

Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command *sudo nano /etc/hostname*

1.1 Use server1 for Server 1

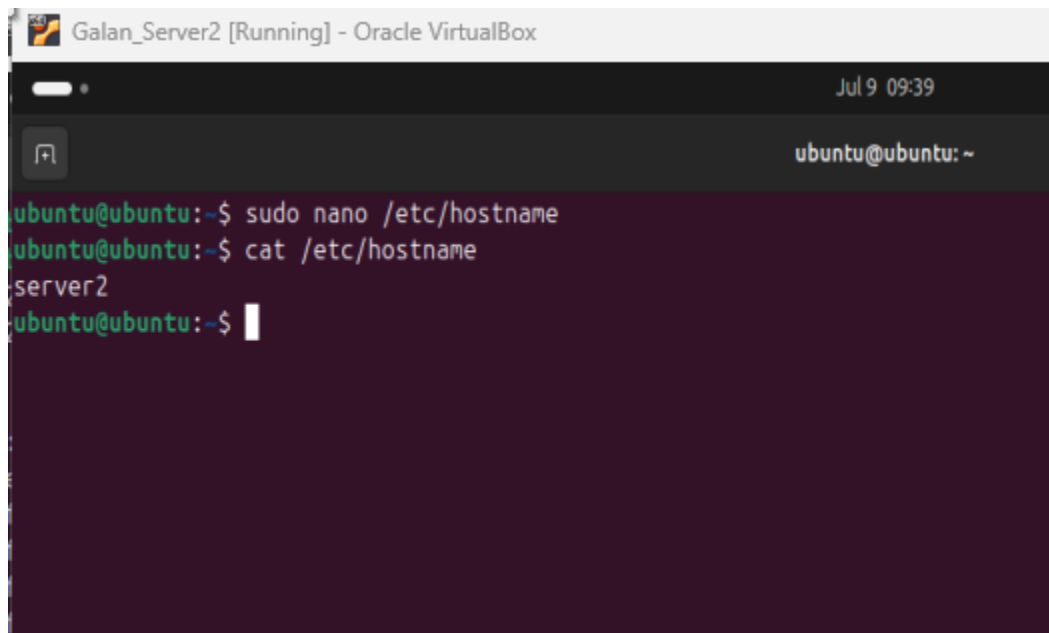


```

Galan_Server1 [Running] - Oracle VirtualBox
Jul 9 09:36
Ubuntu 24.04.2 LTS
ubuntu@ubuntu: ~
Jul 09 08:17:22 ubuntu: sudo nano /etc/hostname
amd64 2.3.1-ubuntu@ubuntu:~$ sudo nano /etc/hostname
Jul 09 08:17:22 ubuntu: cat /etc/hostname
amd64 1:4.4.ubuntu@ubuntu:~$ cat /etc/hostname
Jul 09 08:17:22 ubuntu: server1
amd64 1.4.2-ubuntu@ubuntu:~$
Jul 09 08:17:22 ubuntu:
6.7-2 [2,013
Jul 09 08:17:22 ubuntu:
accessories
Jul 09 08:17:22 ubuntu:
updates/main
Jul 09 08:18:22 ubuntu:
updates/main
Jul 09 08:18:22 ubuntu:

```

1.2 Use server2 for Server 2

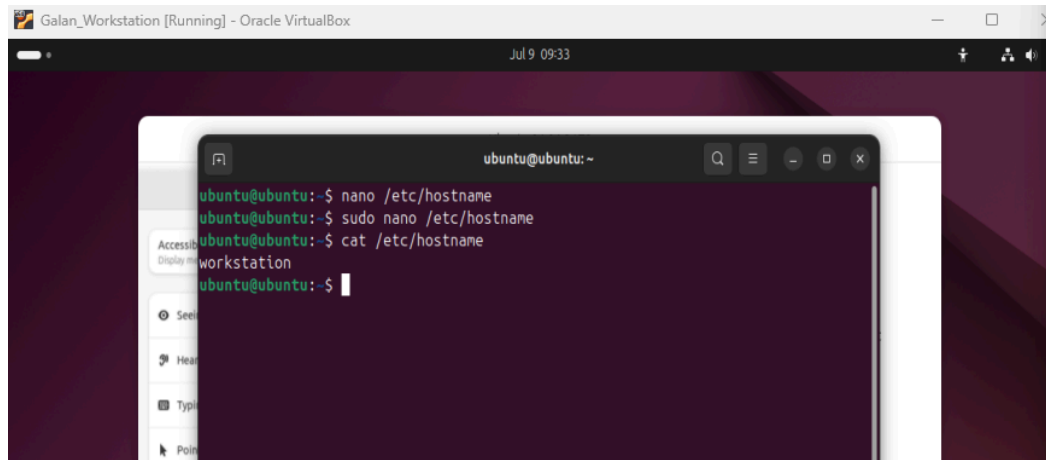


```

Galan_Server2 [Running] - Oracle VirtualBox
Jul 9 09:39
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo nano /etc/hostname
ubuntu@ubuntu:~$ cat /etc/hostname
server2
ubuntu@ubuntu:~$

```

1.3 Use workstation for the Local Machine

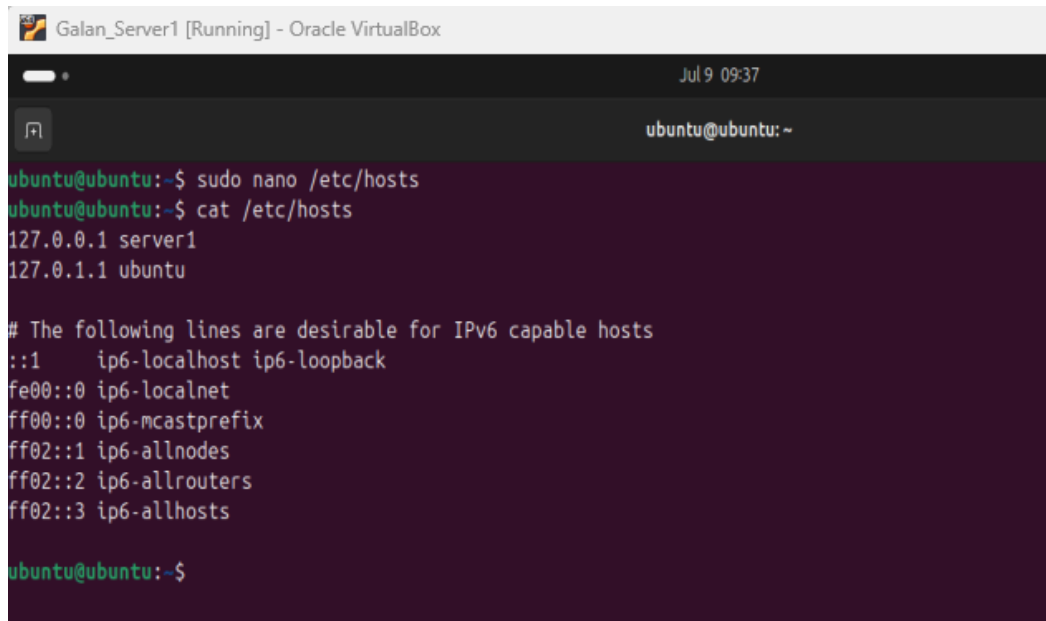


```

Galan_Workstation [Running] - Oracle VirtualBox
Jul 9 09:33
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ nano /etc/hostname
ubuntu@ubuntu:~$ sudo nano /etc/hostname
ubuntu@ubuntu:~$ cat /etc/hostname
workstation
ubuntu@ubuntu:~$

```

2. Edit the hosts using the command *sudo nano /etc/hosts*. Edit the second line.
 - 2.1 Type 127.0.0.1 server 1 for Server 1



```

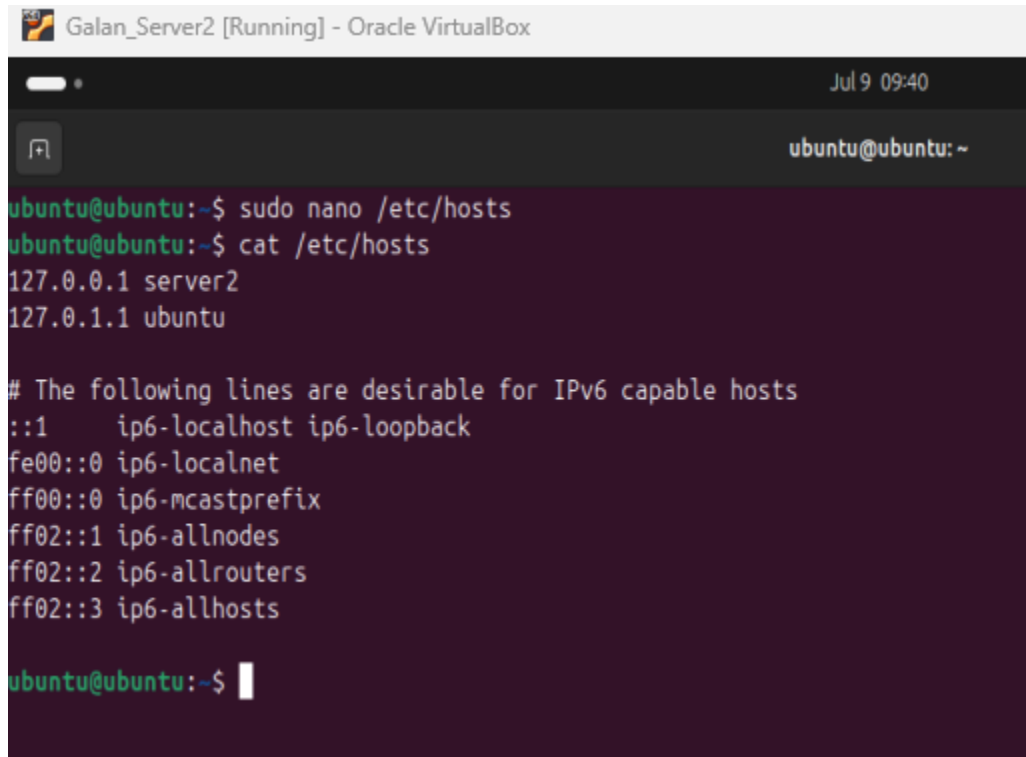
Galan_Server1 [Running] - Oracle VirtualBox
Jul 9 09:37
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo nano /etc/hosts
ubuntu@ubuntu:~$ cat /etc/hosts
127.0.0.1 server1
127.0.1.1 ubuntu

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts

ubuntu@ubuntu:~$

```

- 2.2 Type 127.0.0.1 server 2 for Server 2



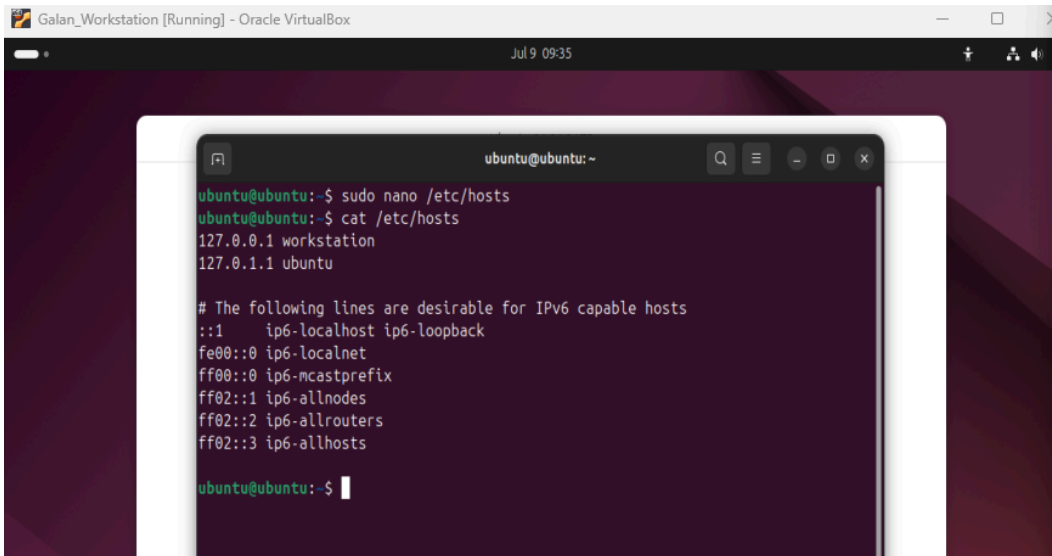
```

ubuntu@ubuntu:~$ sudo nano /etc/hosts
ubuntu@ubuntu:~$ cat /etc/hosts
127.0.0.1 server2
127.0.1.1 ubuntu

# The following lines are desirable for IPv6 capable hosts
::1    ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts

ubuntu@ubuntu:~$
  
```

2.3 Type 127.0.0.1 workstation for the Local Machine



```

ubuntu@ubuntu:~$ sudo nano /etc/hosts
ubuntu@ubuntu:~$ cat /etc/hosts
127.0.0.1 workstation
127.0.1.1 ubuntu

# The following lines are desirable for IPv6 capable hosts
::1    ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts

ubuntu@ubuntu:~$
  
```

Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the command *sudo apt update* and *sudo apt upgrade* respectively.

Workstation

```

Galan_Workstation [Running] - Oracle VirtualBox
Jul 9 09:48
ubuntu@ubuntu: ~

ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [324 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [827 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1079 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [187 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [44.9 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main Icons (48x48) [16.1 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main Icons (64x64) [22.9 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main amd64 Conf Metadata [11.9 kB]

```

```

Galan_Workstation [Running] - Oracle VirtualBox
Jul 9 09:45
ubuntu@ubuntu: ~

ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libllvm20 linux-headers-6.17.0-35-generic linux-hwe-6.17-headers-6.17.0-35
  linux-hwe-6.17-tools-6.17.0-35 linux-image-6.17.0-35-generic
  linux-modules-6.17.0-35-generic linux-modules-extra-6.17.0-35-generic
  linux-tools-6.17.0-35-generic
The following upgrades have been deferred due to phasing:
  fwupd gir1.2-mutter-14 libinput-bin libinput10 libmutter-14-0 mutter-common

```

Server1

Extensions Server1 [Running] - Oracle VirtualBox

Jul 9 09:46

ubuntu@ubuntu: ~

```
ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [515 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1079 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [324 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [827 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [267 kB]
```

Media Galan_Server1 [Running] - Oracle VirtualBox

Jul 9 09:46

ubuntu@ubuntu: ~

```
ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libfwupd3 libllvm20 linux-headers-6.17.0-35-generic linux-hwe-6.17-headers-6.17.0-35
  linux-image-6.17.0-35-generic linux-modules-6.17.0-35-generic linux-modules-extra-6.
  linux-tools-6.17.0-35-generic
```

Server2

```
Galan_Server2 [Running] - Oracle VirtualBox
Jul 9 09:49
ubuntu@ubuntu: ~

ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [324 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1079 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [827 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [515 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [267 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [180 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [187 kB]
```

```
Galan_Server2 [Running] - Oracle VirtualBox
Jul 9 09:49
ubuntu@ubuntu: ~

Reading state information... Done
638 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libllvm20 linux-headers-6.17.0-35-generic linux-hwe-6.17-headers-6.17.0-35 linux-hwe-6.17
  linux-image-6.17.0-35-generic linux-modules-6.17.0-35-generic linux-modules-extra-6.17.0-35
  linux-tools-6.17.0-35-generic
```

2. Install the SSH server using the command *sudo apt install openssh-server*.

Galan_Workstation (Fresh Install) [Running] - Oracle VirtualBox

Jul 11 10:10

```

qimagalan-workstation@qimagalan-workstation-VirtualBox: ~
Setting up mutter-common-bin (46.2-1ubuntu0.24.04.16) ...
Setting up libgtk-4-media-gstreamer (4.14.5+ds-0ubuntu0.10) ...
Setting up gnome-control-center (1:46.7-0ubuntu0.24.04.4) ...
Setting up libmutter-14-0:amd64 (46.2-1ubuntu0.24.04.16) ...
Setting up gir1.2-mutter-14:amd64 (46.2-1ubuntu0.24.04.16) ...
Setting up gnome-shell (46.0-0ubuntu6-24.04.14) ...
Setting up gdm3 (46.2-1ubuntu1-24.04.9) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo apt install opens
sh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 33 not upgraded.
Need to get 832 kB of archives.
After this operation, 6,747 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
  
```

3. Verify if the SSH service has started by issuing the following commands:

3.1 *sudo service ssh start*

3.2 *sudo systemctl status ssh*

Workstation

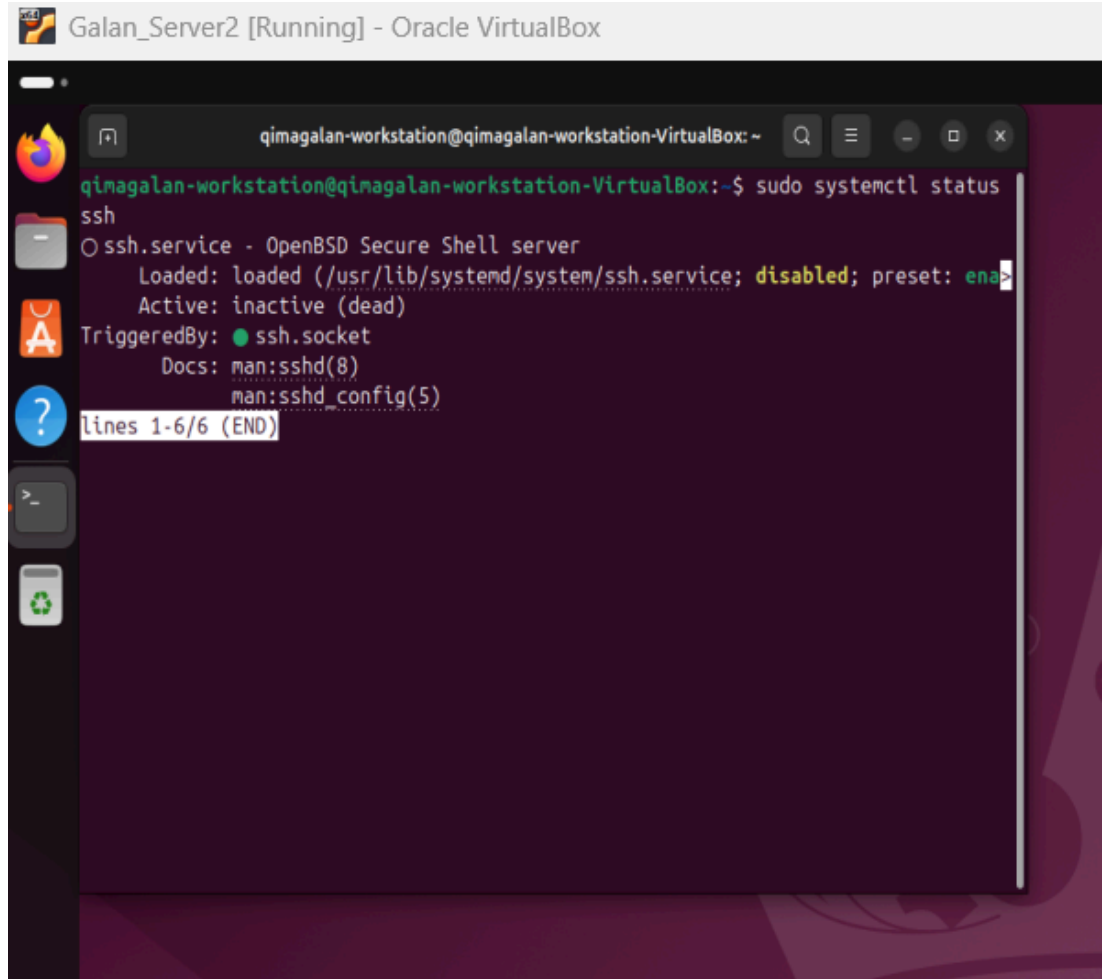
```
qimagalan-workstation@qimagalan-workstation-VirtualBox: ~  
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo service ssh start  
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enable>  
   Active: active (running) since Sat 2026-07-11 10:12:10 PST; 20s ago  
TriggeredBy: ● ssh.socket  
   Docs: man:sshd(8)  
         man:sshd_config(5)  
  Process: 9474 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)  
 Main PID: 9475 (sshd)  
    Tasks: 1 (limit: 4596)  
  Memory: 1.2M (peak: 1.3M)  
     CPU: 51ms  
   CGroup: /system.slice/ssh.service  
           └─9475 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Jul 11 10:12:10 qimagalan-workstation-VirtualBox systemd[1]: Starting ssh.service >  
Jul 11 10:12:10 qimagalan-workstation-VirtualBox sshd[9475]: Server listening on 0>  
Jul 11 10:12:10 qimagalan-workstation-VirtualBox sshd[9475]: Server listening on :>  
Jul 11 10:12:10 qimagalan-workstation-VirtualBox systemd[1]: Started ssh.service ->  
lines 1-18/18 (END)
```

Server1

Galan_Server1 [Running] - Oracle VirtualBox

```
qimagalan-workstation@qimagalan-workstation-VirtualBox: ~  
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo systemctl status  
ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena  
   Active: inactive (dead)  
   TriggeredBy: ● ssh.socket  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$
```

Server2



The screenshot shows a terminal window titled "qimagalan-workstation@qimagalan-workstation-VirtualBox: ~". The user has executed the command `sudo systemctl status ssh`. The output shows that the `ssh.service` is loaded but disabled. The status is "inactive (dead)". The terminal window has a dark background with a sidebar on the left containing icons for a terminal, a file manager, an application store, a help icon, and a trash can.

```
qimagalan-workstation@qimagalan-workstation-VirtualBox: ~  
$ sudo systemctl status ssh  
ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)  
   Active: inactive (dead)  
 TriggeredBy: ● ssh.socket  
    Docs: man:sshd(8)  
          man:sshd_config(5)  
lines 1-6/6 (END)
```

4. Configure the firewall to all port 22 by issuing the following commands:

4.1 `sudo ufw allow ssh`

4.2 `sudo ufw enable`

4.3 `sudo ufw status`

Workstation

```
Galan_Workstation (Fresh Install) [Running] - Oracle VirtualBox
qimagalan-workstation@qimagalan-workstation-VirtualBox: ~
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo ufw enabled
ERROR: Invalid syntax

Usage: ufw COMMAND

Commands:
enable                enables the firewall
disable              disables the firewall
default ARG           set default policy
logging LEVEL        set logging to LEVEL
allow ARGS           add allow rule
deny ARGS            add deny rule
reject ARGS          add reject rule
limit ARGS           add limit rule
delete RULE|NUM      delete RULE
insert NUM RULE       insert RULE at NUM
prepend RULE         prepend RULE
route RULE           add route RULE
route delete RULE|NUM delete route RULE
route insert NUM RULE insert route RULE at NUM
```

```
Galan_Workstation (Fresh Install) [Running] - Oracle VirtualBox
qimagalan-workstation@qimagalan-workstation-VirtualBox: ~
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo ufw status
Status: inactive
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo ufw status
Status: active

To           Action From
--           -
22/tcp       ALLOW  Anywhere
22/tcp (v6)  ALLOW  Anywhere (v6)

qimagalan-workstation@qimagalan-workstation-VirtualBox:~$
```

Server1

Galan_Server1 [Running] - Oracle VirtualBox

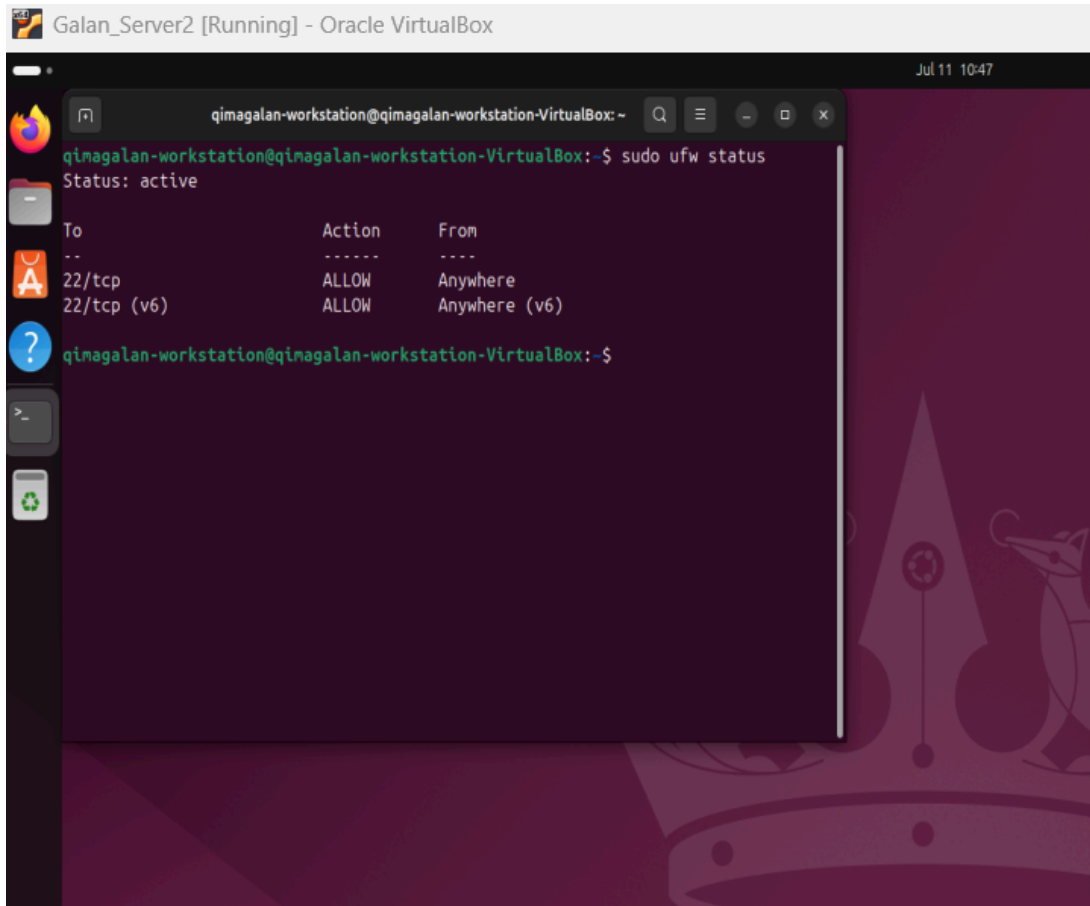
Jul 11 10:43

```
qimagalan-workstation@qimagalan-workstation-VirtualBox: ~
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo systemctl status
ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena>
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
         man:sshd_config(5)
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

qimagalan-workstation@qimagalan-workstation-VirtualBox:~$
```

Server2



```
qimagalan-workstation@qimagalan-workstation-VirtualBox: ~  
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$ sudo ufw status  
Status: active  

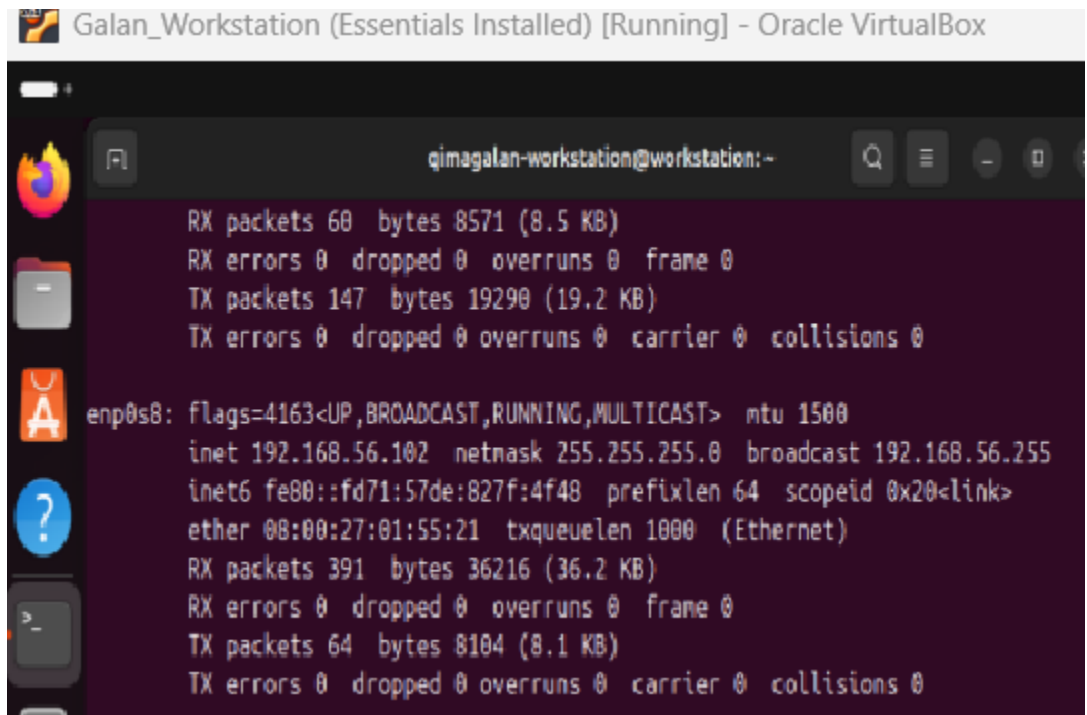

| To          | Action | From          |
|-------------|--------|---------------|
| --          | ----   | ----          |
| 22/tcp      | ALLOW  | Anywhere      |
| 22/tcp (v6) | ALLOW  | Anywhere (v6) |

  
qimagalan-workstation@qimagalan-workstation-VirtualBox:~$
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command *ifconfig* and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.

1.1 Server 1 IP address: 192.168.56.102

A screenshot of a terminal window titled 'Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox'. The terminal shows the output of the 'ifconfig' command for the 'enp0s8' interface. The output includes statistics for RX and TX packets and bytes, as well as interface flags, MTU, and IP addresses. The IP address 192.168.56.104 is highlighted in the original image.

```
qimagalan-workstation@workstation:~  
RX packets 60 bytes 8571 (8.5 KB)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 147 bytes 19290 (19.2 KB)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255  
inet6 fe80::fd71:57de:827f:4f48 prefixlen 64 scopeid 0x20<link>  
ether 08:00:27:01:55:21 txqueuelen 1000 (Ethernet)  
RX packets 391 bytes 36216 (36.2 KB)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 64 bytes 8104 (8.1 KB)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

1.2 Server 2 IP address: 192.168.56.104

Galan_Server1 (Fresh Install) [Running] - Oracle VirtualBox

```

qimagalan-workstation@server1: ~
RX packets 91 bytes 22133 (22.1 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 186 bytes 26177 (26.1 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
inet6 fe80::1799:246d:9afb:7d2d prefixlen 64 scopeid 0x20<link>
ether 08:00:27:da:18:53 txqueuelen 1000 (Ethernet)
RX packets 590 bytes 59862 (59.8 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 64 bytes 8952 (8.9 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 162 bytes 14326 (14.3 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 162 bytes 14326 (14.3 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

qimagalan-workstation@server1:~$
  
```

1.3 Server 3 IP address: 192.168.56.106

Galan_Server2_Clone [Running] - Oracle VirtualBox

```

qimagalan-workstation@server2: ~
RX packets 87 bytes 20246 (20.2 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 172 bytes 22761 (22.7 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.56.106 netmask 255.255.255.0 broadcast 192.168.56.255
inet6 fe80::a8a6:1ced:75cc:35e8 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:71:3e:00 txqueuelen 1000 (Ethernet)
RX packets 348 bytes 32306 (32.3 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 63 bytes 8087 (8.0 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 160 bytes 14160 (14.1 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 160 bytes 14160 (14.1 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

qimagalan-workstation@server2:~$
  
```

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: ☒ Successful ☐ Not Successful

Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

```

qimagalan-workstation@workstation: ~
qimagalan-workstation@workstation:~$ ping -c 4 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data:
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=2.42 ns
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=1.10 ns
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=1.34 ns
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=1.89 ns

--- 192.168.56.104 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3582ms
rtt min/avg/max/mdev = 1.095/1.686/2.416/0.510 ns
qimagalan-workstation@workstation:~$
  
```

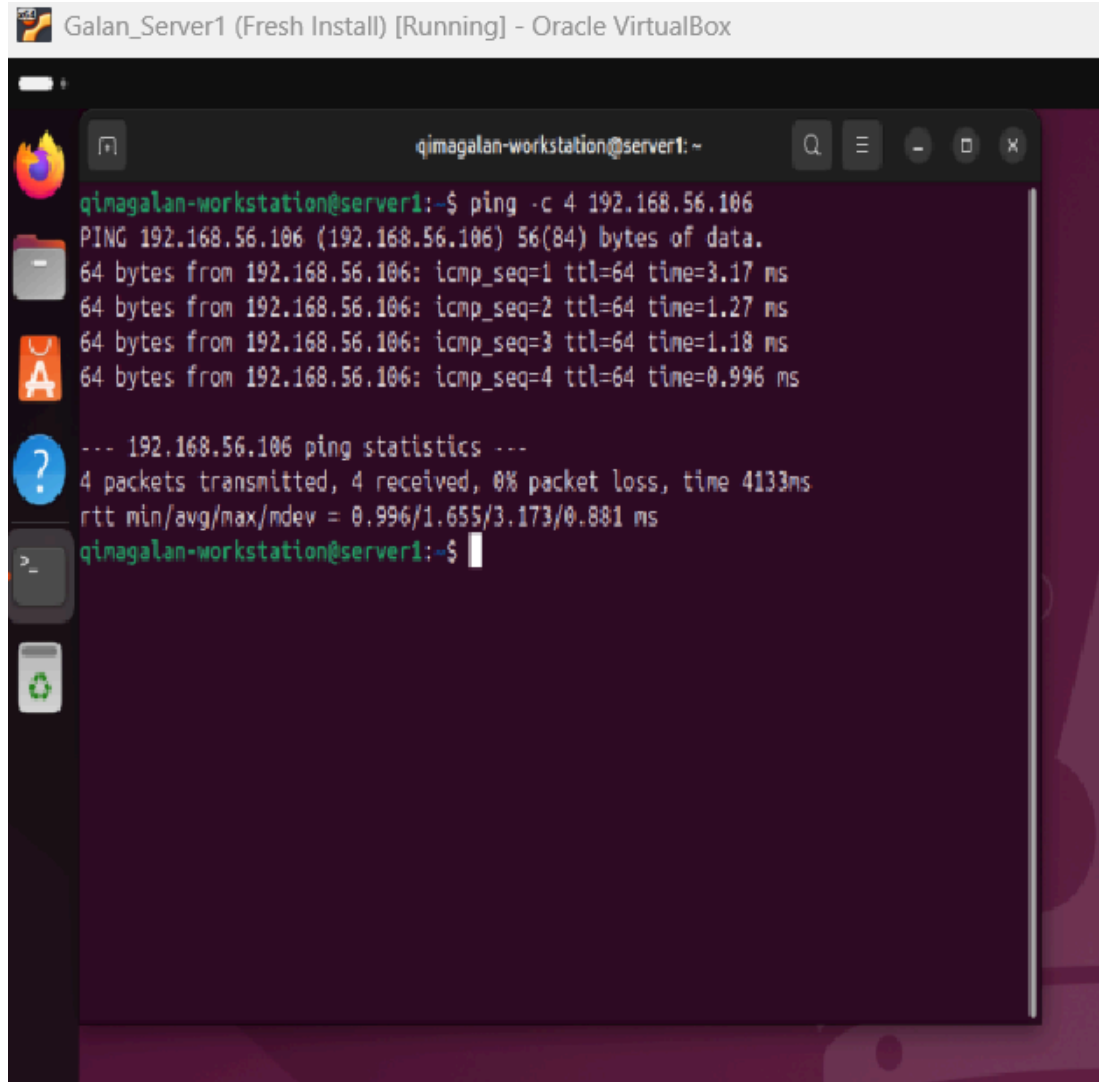
2.2 Connectivity test for Local Machine 1 to Server 2: ☒ Successful ☐ Not Successful

Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

```
qimagalan-workstation@workstation:~$ ping -c 4 192.168.56.106
PING 192.168.56.106 (192.168.56.106) 56(84) bytes of data:
64 bytes from 192.168.56.106: icmp_seq=1 ttl=64 time=3.16 ms
64 bytes from 192.168.56.106: icmp_seq=2 ttl=64 time=1.27 ms
64 bytes from 192.168.56.106: icmp_seq=3 ttl=64 time=1.23 ms
64 bytes from 192.168.56.106: icmp_seq=4 ttl=64 time=1.32 ms

--- 192.168.56.106 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.233/1.746/3.158/0.815 ms
qimagalan-workstation@workstation:~$
```

2.3 Connectivity test for Server 1 to Server 2: ☒ Successful ☐ Not Successful



Galan_Server1 (Fresh Install) [Running] - Oracle VirtualBox

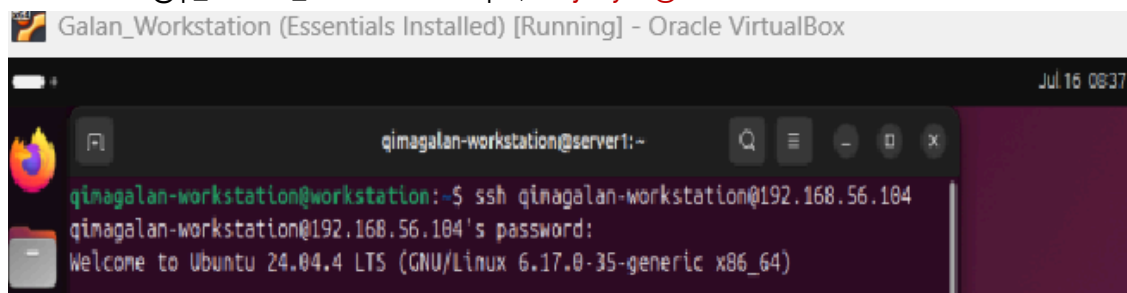
```
qimagalan-workstation@server1:~$ ping -c 4 192.168.56.106
PING 192.168.56.106 (192.168.56.106) 56(84) bytes of data:
64 bytes from 192.168.56.106: icmp_seq=1 ttl=64 time=3.17 ms
64 bytes from 192.168.56.106: icmp_seq=2 ttl=64 time=1.27 ms
64 bytes from 192.168.56.106: icmp_seq=3 ttl=64 time=1.18 ms
64 bytes from 192.168.56.106: icmp_seq=4 ttl=64 time=0.996 ms

--- 192.168.56.106 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 4133ms
rtt min/avg/max/mdev = 0.996/1.655/3.173/0.881 ms
qimagalan-workstation@server1:~$
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

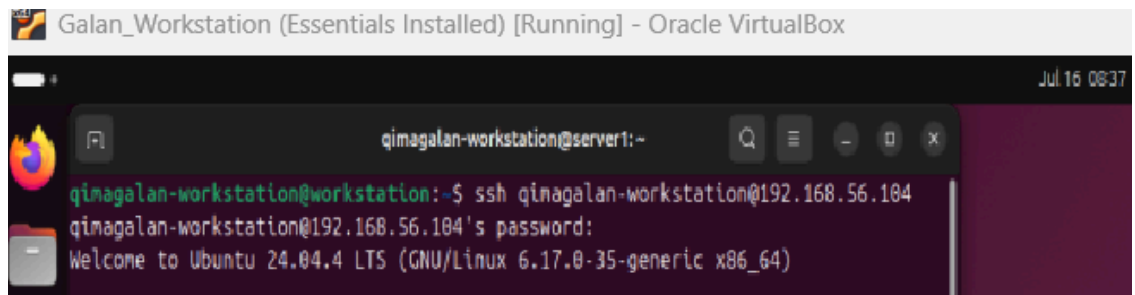
1.1 `ssh username@ip_address_server1` for example, `ssh jvtaylor@192.168.56.120`



Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

```
qimagalan-workstation@server1:~$ ssh qimagalan-workstation@192.168.56.104
qimagalan-workstation@192.168.56.104's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)
```

1.2 Enter the password for server 1 when prompted

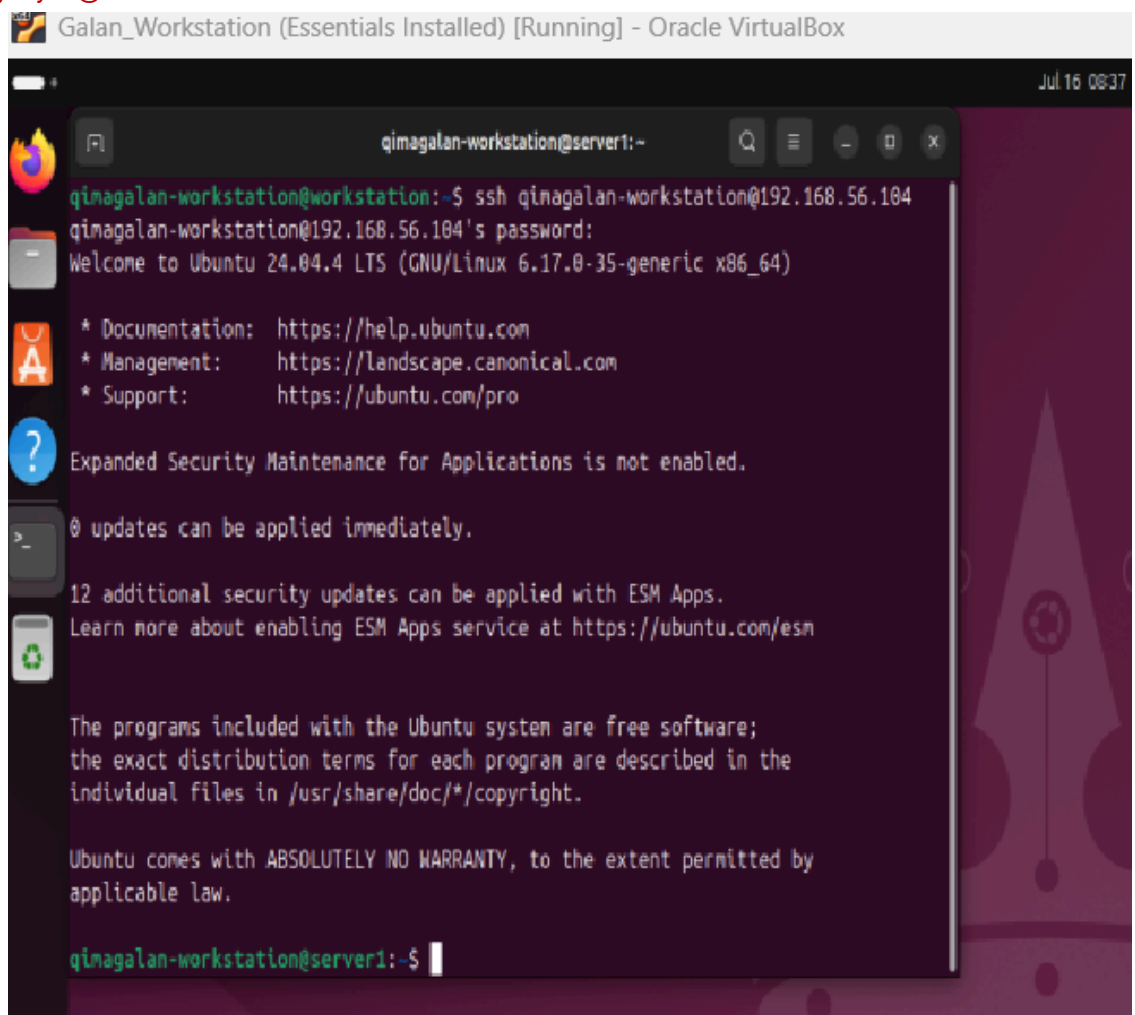


Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

```

qimagalan-workstation@server1:~
qimagalan-workstation@workstation:~$ ssh qimagalan-workstation@192.168.56.104
qimagalan-workstation@192.168.56.104's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)
  
```

- 1.3 Verify that you are in server 1. The user should be in this format user@server1. For example, *jvtaylor@server1*



Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

```

qimagalan-workstation@server1:~
qimagalan-workstation@workstation:~$ ssh qimagalan-workstation@192.168.56.104
qimagalan-workstation@192.168.56.104's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

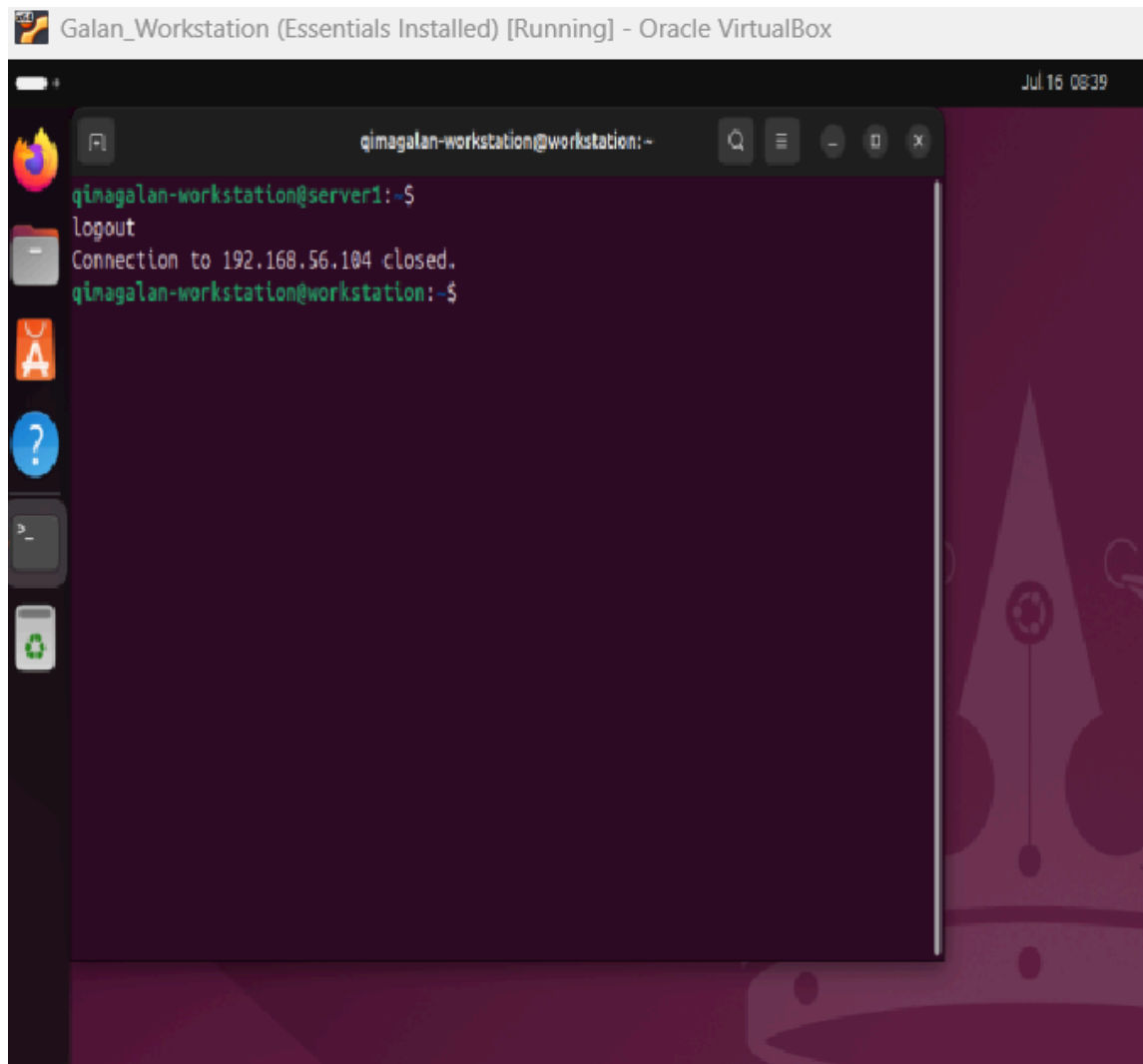
12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

qimagalan-workstation@server1:~$
  
```

2. Logout of Server 1 by issuing the command *control + D*.



3. Do the same for Server 2.

Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

Jul 16 08:41

qinagalan-workstation@server2: ~

```
qinagalan-workstation@workstation:~$ ssh qinagalan-workstation@192.168.56.106
The authenticity of host '192.168.56.106 (192.168.56.106)' can't be established.
ED25519 key fingerprint is SHA256:CDjezw3sGvTyk/pOWS40HdDwiov6Ig6eLvLzHyvJ1y4.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.106' (ED25519) to the list of known hosts.
qinagalan-workstation@192.168.56.106's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

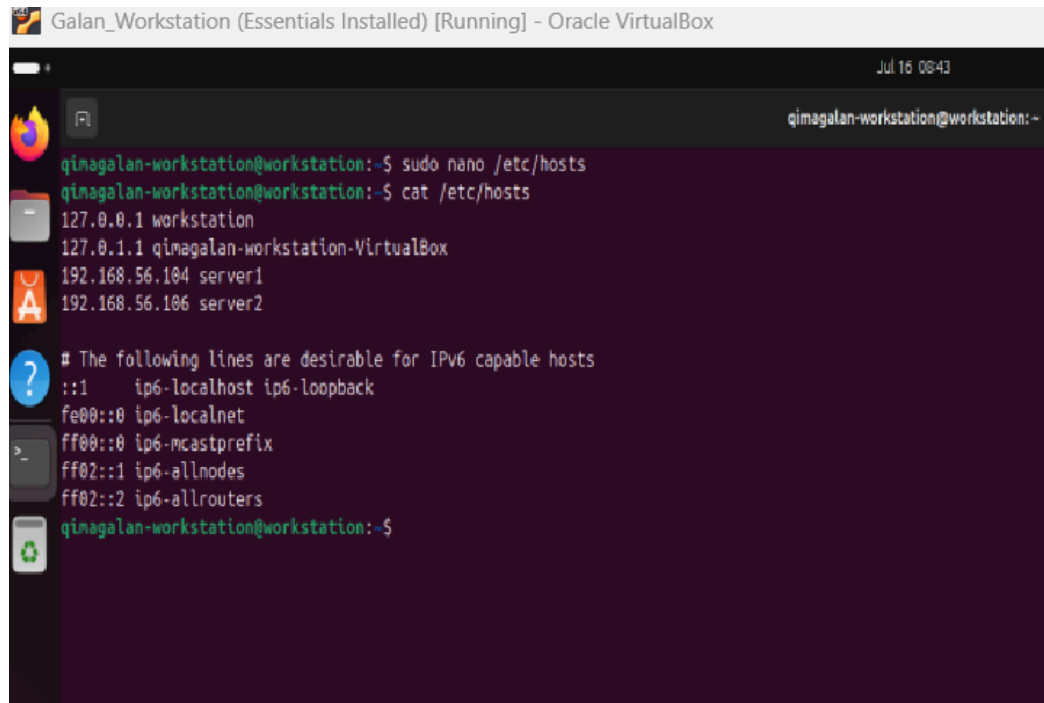
12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

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Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

qinagalan-workstation@server2:~$
```

4. Edit the hosts of the Local Machine by issuing the command `sudo nano /etc/hosts`. Below all texts type the following:
 - 4.1 `IP_address server 1` (provide the ip address of server 1 followed by the hostname)
 - 4.2 `IP_address server 2` (provide the ip address of server 2 followed by the hostname)
 - 4.3 Save the file and exit.



Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

```
qimagalan-workstation@workstation:~$ cat /etc/hosts
127.0.0.1 workstation
127.0.1.1 qimagalan-workstation-VirtualBox
192.168.56.104 server1
192.168.56.106 server2

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
qimagalan-workstation@workstation:~$
```

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do `ssh jvtaylor@server1`. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

Workstation (Local Machine) -> Server 1

Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

Jul 16 08:44

qimagalan-workstation@server1:~

```
qimagalan-workstation@workstation:~$ ssh qimagalan-workstation@server1
The authenticity of host 'server1 (192.168.56.104)' can't be established.
ED25519 key fingerprint is SHA256:CDjezw3sGvTyk/pOWS40HdDwIov6Ig6eLvLzHyvJ1y4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
  ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
qimagalan-workstation@server1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Jul 16 08:39:24 2026 from 192.168.56.102
qimagalan-workstation@server1:~$
```

Workstation (Local Machine) -> Server 2

Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

```

qinagalan-workstation@workstation:~$ ssh qinagalan-workstation@server2
The authenticity of host 'server2 (192.168.56.106)' can't be established.
ED25519 key fingerprint is SHA256:CDjezw3sGvTyk/pOWS40HdDwiov6Ig6eLvLzHyvJ1y4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.
qinagalan-workstation@server2's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Jul 16 08:40:33 2026 from 192.168.56.102
qinagalan-workstation@server2:~$
  
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?

Linux maintains everything in its filesystem, even the configurations. As I also observe from task 1 above, that if we would like the efficiency of simply using hostname instead of an IP address for ssh connections, there are 2 files we also need to modify. The `/etc/hostname` configures the hostname for the current machine, and then `/etc/hosts` simply maps 127.0.0.1 (localhost) to the hostname. Above, we will see that server1, server2, and workstation is mapped into the respective local hosts of each machine so that we could distinguish the hostnames between them when connecting in ssh, because if we have not modified `/etc/hostname` and `/etc/hosts` the machines will be named localhost by default and might cause errors.

2. How secured is SSH?

SSH is an upgrade compared to its predecessor, the Telnet. Telnet uses plain text to send data over the connection of server and client meanwhile SSH makes use of an encryption technology to prevent reading and changes on the message sent between server and client. It is able to establish secure connections by creating public and private key pairs, since this allows the server to determine legitimate connection requests in which only the correct private key could pair with its public key (Oloid Desk, 2026), and it also does this discreetly. It also establishes a session key that allows encryption-decryption of messages between the server and remote client. Default configurations of ssh however still needs to be further tinkered, such as adding passphrases for the keys, as well as relocating ssh to a different port as compared to its default port 22 which could become vulnerable to bruteforce attacks.

OLOID Desk. (2026, May 7). *Everything you must know about security shell (SSH)*. OLOID. <https://www.oloid.com/blog/everything-you-must-know-about-security-shell>

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Chatgpt

Purpose of Use:

Reference Citation Formatting only

Extent of Use:

Answers in the reflection are all mine and only the citation format was assisted by chatgpt in the reflection portion.

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